

GridSFM as an AC Power-Flow Surrogate

Exact formulation of the released backbone and PPC adaptation

Implementation note for `train_valid_test_gridsfm.py`

August 2026

Scope. These equations describe the released `gridsfm.GridTransformerBackbone` driven by the AC-PF branch-row adapter. GridSFM was originally structured for AC-OPF and ships pretrained weights; here its voltage heads are fine-tuned or retrained against Newton PF labels. Optional known-operator postprocessing is excluded.

1 Target AC-PF Mapping

For $\underline{V}_i = v_i e^{j\theta_i}$, define

$$\underline{\mathbf{S}}^{\text{calc}} = \underline{\mathbf{V}} \odot \overline{\mathbf{Y}\underline{\mathbf{V}}}, \quad \Delta \mathbf{P} = \mathbf{P}^{\text{set}} - \Re \underline{\mathbf{S}}^{\text{calc}}, \quad \Delta \mathbf{Q} = \mathbf{Q}^{\text{set}} - \Im \underline{\mathbf{S}}^{\text{calc}}. \quad (1)$$

The surrogate learns

$$f_{\Theta}(\mathcal{G}, \underline{\mathbf{S}}^{\text{set}}, \mathbf{V}^0) \mapsto (\hat{\mathbf{v}}, \hat{\boldsymbol{\theta}}) \approx (\mathbf{v}^*, \boldsymbol{\theta}^*). \quad (2)$$

Active residuals are evaluated on non-slack buses and reactive residuals on PQ buses, matching the fixed-bus-type Newton system.

2 PPC-to-GridSFM Heterogeneous Graph

Before preprocessing, the adapter creates node sets

$$\mathcal{V}_b, \quad \tilde{\mathcal{V}}_g, \quad \mathcal{V}_d, \quad \mathcal{V}_s, \quad (3)$$

for buses, proxy generators, loads, and shunts. The released preprocessing then promotes branches to nodes and introduces cycle nodes:

$$\mathcal{V} = \mathcal{V}_b \cup \tilde{\mathcal{V}}_g \cup \mathcal{V}_d \cup \mathcal{V}_s \cup \mathcal{V}_{\ell} \cup \mathcal{V}_{\tau} \cup \mathcal{V}_c. \quad (4)$$

Here $\mathcal{V}_{\ell}$ and $\mathcal{V}_{\tau}$ represent AC-line and transformer branch rows, while $\mathcal{V}_c$ represents a fundamental cycle basis. Native GridSFM expects physical generator rows; the PPC PF adapter instead creates one proxy generator per PV/REF bus.

2.1 Raw node features

For a bus i ,

$$\mathbf{x}_i^b = [V_{n,i}^{\text{kV}}, t_i, v_i^{\min}, v_i^{\max}] \in \mathbb{R}^4, \quad (5)$$

where $t_i \in \{1, 2, 3, 4\}$ encodes PQ, PV, REF, or isolated. The start magnitude is retained separately as $v_i^{\text{set}} = v_i^0$.

For proxy generator $\tilde{g}_i$, load d_i , and shunt s_i ,

$$\mathbf{x}_{\tilde{g}_i}^g = [m_{\text{base}}, P_g, P_g^{\min}, P_g^{\max}, Q_g, Q_g^{\min}, Q_g^{\max}, v_g^{\text{set}}, c_2, c_1, c_0] \in \mathbb{R}^{11}, \quad (6)$$

$$\mathbf{x}_{d_i}^d = [P_{d,i}, Q_{d,i}] \in \mathbb{R}^2, \quad (7)$$

$$\mathbf{x}_{s_i}^s = [B_i^{\text{sh}}, G_i^{\text{sh}}] \in \mathbb{R}^2. \quad (8)$$

In the PF adapter, generator limits are synthesized around bus-level injections, costs are zero, and $v_g^{\text{set}} = v_i^0$. Reactive injection is set to zero at PV/REF buses before graph construction because it is not specified by fixed-type PF. Load channels are formed from the negative net injection and clipped at zero.

2.2 Line and transformer features

The adapter divides live branches into AC lines and transformers. An AC-line row is

$$\mathbf{e}_{ij}^\ell = [\theta^{\min}, \theta^{\max}, b_{fr}, b_{to}, r, x, S_A, S_B, S_C] \in \mathbb{R}^9, \quad (9)$$

and a transformer row is

$$\mathbf{e}_{ij}^\tau = [\theta^{\min}, \theta^{\max}, r, x, S_A, S_B, S_C, \tau, \phi, b_{fr}, b_{to}] \in \mathbb{R}^{11}. \quad (10)$$

The series impedance is recovered from the PPC admittance,

$$r + jx = 1/y_{ij}, \quad (11)$$

and transformers are identified from metadata, non-unit tap, phase shift, and optionally endpoint nominal-voltage mismatch.

3 Branch Promotion and Cycle Representation

Each branch row becomes a node whose initial feature is its line or transformer attribute. For branch $e = (i, j)$, signed endpoint edges are created:

$$i \xrightarrow{+1} e, \quad j \xleftarrow{-1} e. \quad (12)$$

Let B_1 denote this oriented bus-branch incidence. A fundamental cycle basis is built from a spanning tree. A cycle c has features

$$\mathbf{x}_c^c = [|c|, \sum_{e \in c} |x_e|, \sum_{e \in c} |r_e|, \min_{e \in c} S_e^{\max}] \in \mathbb{R}^4, \quad (13)$$

and signed cycle-branch incidence $B_2(c, e) \in \{-1, 0, +1\}$. This creates the sequence

$$\text{bus (0-form)} \longleftrightarrow \text{branch (1-form)} \longleftrightarrow \text{cycle (2-form)}. \quad (14)$$

4 Static Physics and Positional Features

4.1 Laplacian/DC features

Using branch susceptances $b_{ij} \approx 1/x_{ij}$, GridSFM builds

$$L_B = \sum_{(i,j)} b_{ij} (\mathbf{e}_i - \mathbf{e}_j)(\mathbf{e}_i - \mathbf{e}_j)^\top. \quad (15)$$

A reduced DC solve supplies

$$L_{B, \setminus r} \boldsymbol{\theta}_{\text{DC}} = \mathbf{P}_{\setminus r}, \quad (16)$$

with the slack balancing total injection. Bus DC features are

$$\mathbf{p}_i^{\text{DC}} = [\theta_i^{\text{DC}}, \text{mean}_{j \in \mathcal{N}(i)} |\theta_i^{\text{DC}} - \theta_j^{\text{DC}} - \phi_{ij}|, P_i^{\text{raw}}] \in \mathbb{R}^3. \quad (17)$$

Each branch receives

$$[\Delta \theta_{ij}^{\text{DC}}, P_{ij}^{\text{DC}}, \min(|P_{ij}^{\text{DC}}|/S_{ij}^{\max}, 2), \text{sign } P_{ij}^{\text{DC}}] \in \mathbb{R}^4. \quad (18)$$

Five landmark effective-resistance moments are added to each bus. A four-column bus-type one-hot is appended, giving $4 + 5 + 3 + 4 = 16$ bus columns before learned Hodge encoding.

4.2 Learned Hodge positional encoding

For each active form $q \in \{0, 1, 2\}$, let L_q be the signed Hodge operator induced by B_1 and B_2 :

$$L_0 = B_1 B_1^\top, \quad L_1 = B_1^\top B_1 + B_2^\top B_2, \quad L_2 = B_2 B_2^\top. \quad (19)$$

After projecting to $K_H = 8$ channels, GridSFM performs $T_H = 4$ learned diffusion steps:

$$\mathbf{z}_q^{(0)} = W_q \mathbf{x}_q, \quad \mathbf{z}_q^{(t+1)} = \mathbf{z}_q^{(t)} - \boldsymbol{\alpha}_{q,t} \odot L_q \mathbf{z}_q^{(t)}. \quad (20)$$

The concatenated trajectory is mixed and normalized:

$$\mathbf{p}_q^{\text{Hodge}} = \text{LN}\left(\text{MLP}_q[\mathbf{z}_q^{(0)} \parallel \dots \parallel \mathbf{z}_q^{(T_H)}]\right). \quad (21)$$

An eight-dimensional graph context also summarizes system size, load/capacity ratios, voltage-band width, ratings, and admittance scales:

$$\mathbf{g}_{\text{ctx}} \in \mathbb{R}^8. \quad (22)$$

5 GridTransformer Blocks

Each node type is independently normalized and projected to hidden dimension $d = 128$. The released default has $L = 8$ blocks and $H = 4$ heads.

5.1 Per-type linear self-attention

For each node type t , attention is global within each graph but only among nodes of that same type. With feature map $\varphi(x) = \text{ELU}(x) + 1$,

$$Q = W_Q h, \quad K = W_K h, \quad Z = W_V h, \quad (23)$$

$$\mathbf{M}_{\gamma, h} = \sum_{j \in \gamma, t} \varphi(K_j^h) (Z_j^h)^\top, \quad \mathbf{k}_{\gamma, h} = \sum_{j \in \gamma, t} \varphi(K_j^h), \quad (24)$$

$$\text{Attn}_{t, h}(i) = \frac{\varphi(Q_i^h)^\top \mathbf{M}_{\gamma(i), h}}{\varphi(Q_i^h)^\top \mathbf{k}_{\gamma(i), h} + \epsilon}. \quad (25)$$

This Performer-style expression is linear in the number of nodes rather than dense quadratic attention. The residual update is

$$h_t \leftarrow h_t + W_O \parallel_{h=1}^H \text{Attn}_{t, h}. \quad (26)$$

5.2 Typed local message passing and FFN

For signed incidence relations, GridSFM uses

$$m_{u \rightarrow v} = s_{uv} W_{\text{msg}}^{(r)} h_u, \quad \tilde{h}_v = W_{\text{self}}^{(r)} h_v + \text{mean}_{u \in \mathcal{N}_r(v)} m_{u \rightarrow v}. \quad (27)$$

For generator/load/shunt association relations it uses relation-specific GraphSAGE mean aggregation. Relation outputs with the same target type are summed by HeteroConv:

$$h_v \leftarrow h_v + \sum_{r: \text{dst}(r)=t(v)} \tilde{h}_v^{(r)}. \quad (28)$$

Each block closes with a pre-normalized residual feed-forward network:

$$h_v \leftarrow h_v + W_2^{t(v)} \text{GELU}(W_1^{t(v)} \text{LN}(h_v)). \quad (29)$$

6 Fusion and Terminal Heads

After all blocks, relation-aware mean aggregates are fused into bus embeddings:

$$h_i^{\text{grid}, b} = \text{LN}(W_b h_i^b + W_\ell \bar{h}_i^\ell + W_\tau \bar{h}_i^\tau + W_g \bar{h}_i^g + W_d \bar{h}_i^d + W_s \bar{h}_i^s + W_c \bar{h}_i^c). \quad (30)$$

Generator embeddings fuse their associated bus. A global embedding is obtained from mean and maximum pools of buses, line branches, transformer branches, and cycles.

GridSFM applies its heads once, after fusion. Generator predictions are bounded around their intervals:

$$\hat{P}_g = \text{clip}_{[P_g^{\min}, P_g^{\max}]} \left(\frac{1}{2} (P_g^{\min} + P_g^{\max}) + H_P(h_g) \right), \quad (31)$$

$$\hat{Q}_g = \text{clip}_{[Q_g^{\min}, Q_g^{\max}]} \left(\frac{1}{2} (Q_g^{\min} + Q_g^{\max}) + H_Q(h_g) \right). \quad (32)$$

Voltage-angle head. The prediction combines a cached DC prior with a learned residual:

$$\hat{\theta}_i = \text{anchor} \left[\pi \tanh(\theta_i^{\text{DC}}/\pi) + s_\theta \tanh(H_\theta(h_i)/s_\theta) \right]. \quad (33)$$

The magnitude head receives the start/setpoint magnitude explicitly:

$$\hat{v}_i = \text{clip}_{[v_i^{\min}, v_i^{\max}]} H_v([h_i \| v_i^{\text{set}}]). \quad (34)$$

Finally, exact π -model formulas compute predicted line/transformer flows from $(\hat{v}, \hat{\theta}, \hat{P}_g, \hat{Q}_g)$, and a separate feasibility head uses 24 pooled stress features. The AC-PF adapter consumes only $(\hat{v}, \hat{\theta})$.

Gradient caveat. The DC prior is computed from $\hat{P}_g$ through a detached CPU/SciPy path. Under the default voltage-only PF loss, the generator head therefore receives no gradient through the angle prior. Its pretrained behavior is retained, but PF fine-tuning updates it only if a separate generator-head loss is enabled.

7 AC-PF Training Loss and Metrics

The voltage-state loss is

$$\mathcal{L}_{\text{state}} = \frac{1}{N} \sum_i \left[(\hat{v}_i - v_i^*)^2 + \text{wrap}(\hat{\theta}_i - \theta_i^*)^2 \right]. \quad (35)$$

With $\mathcal{M}_P = \mathcal{V} \setminus \mathcal{V}_{\text{REF}}$ and $\mathcal{M}_Q = \mathcal{V}_{\text{PQ}}$,

$$\mathcal{L}_{\text{phys}} = \frac{1}{|\mathcal{M}_P| + |\mathcal{M}_Q|} \left[\sum_{i \in \mathcal{M}_P} \rho(\Delta P_i) + \sum_{i \in \mathcal{M}_Q} \rho(\Delta Q_i) \right], \quad (36)$$

where $\rho(r) = \log \cosh(\text{clip}(r, -30, 30))$ by default. The training objective is

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{state}} + 10^{-2} \mathcal{L}_{\text{phys}}.} \quad (37)$$

The script computes training physics through complex64 tensors but uses the complex128 dataset/Ybus path for reported validation and test residual metrics.

8 Interpretive Summary

Property	GridSFM PF adaptation
Graph	Seven node types: bus, proxy generator, load, shunt, AC branch, transformer branch, and cycle.
Topology encoding	Signed incidence, fundamental cycles, static Laplacian/DC features, and learned Hodge diffusion.
Neural core	Eight blocks of per-type linear attention, typed local GNN aggregation, and residual FFNs.
Voltage decoder	One terminal head; angle is DC prior plus learned residual, magnitude is bounded.
Physics inside forward	DC prior, exact branch-flow calculation, and terminal stress/feasibility features; no iterative AC residual feedback.
Physics outside forward	Exact-Ybus robust AC-PF loss and complex128 evaluation metrics.
Pretraining	Released GridSFM checkpoint may be loaded and fine-tuned.

Code provenance

The formulation follows `gridsfm/model.py`, `gridsfm/blocks.py`, `gridsfm/cycle_basis.py`, `gridsfm/hodge_pe.py`, `gridsfm/pe_features.py`, `gridsfm/dc_prior.py`, and the AC-PF adapter in `train_valid_test_gridsfm.py`, as present in August 2026.